

WATERSHED WATER QUALITY INTELLIGENCE DASHBOARD

Data-Driven Environmental
Risk Analysis using *Power BI*



Author

R. R. MAO



Project Type

Environmental Data Analytics Portfolio



Tools

Power BI | DAX | Power Query



Dataset

Water Quality Monitoring Data



Year

2026



AVG PHOSPHORUS
(mg/L)

3.47

● MODERATE



AVG DISSOLVED
OXYGEN (mg/L)

7.71

● GOOD



WATER QUALITY
INDEX (WQI)

72

● MODERATE



POLLUTION
RISK LEVEL

MODERATE

● MEDIUM RISK



INSIGHT • INNOVATION • SUSTAINABILITY



■ ABSTRACT

This project presents a data-driven analysis of watershed water quality using Power BI. It investigates how rainfall, land use, and pollution events influence key environmental indicators including turbidity, nutrient concentration, and fish population health. The analysis identifies agricultural land use and rainfall-driven runoff as the primary contributors to water quality degradation. The dashboard enables interactive exploration and supports environmental decision-making and watershed management strategies.

■ INTRODUCTION

Water quality degradation is a critical environmental issue influenced by both natural processes and human activities.

This study aims to analyze spatial and temporal patterns in water quality and identify key environmental drivers using interactive data visualization techniques.

The focus is on three core dimensions:

- Hydrological influence (Rainfall)
 - Land-use impact (Agriculture, Urban, Forest)
 - Ecological response (Fish population dynamics)
-

■ DATA SOURCE

The dataset consists of environmental monitoring data collected across multiple stations.

It includes:

- Water quality indicators (Phosphorus, Dissolved Oxygen, Turbidity)
 - Rainfall data
 - Land use classification
 - Fish population metrics
 - Pollution event indicators
-

■ DATA PREPROCESSING

Data preprocessing was performed using Power Query in Power BI.

Steps included:

- Removal of missing and inconsistent values
 - Standardization of categorical variables (Land Use, Station ID)
 - Date formatting normalization
 - Outlier review for environmental variables
 - Consolidation into a single analytical table (WaterQuality)
-

■ METHODOLOGY

1 Analytical Framework

The analysis follows a multi-factor environmental systems approach:

- Rainfall → Turbidity (hydrological impact)
 - Land Use → Nutrient Pollution (anthropogenic impact)
 - Pollution Events → Fish Population (ecological impact)
-

2 Statistical Approach

- Descriptive statistics (mean values)
 - Comparative analysis across land-use categories
 - Correlation-based visual exploration
-

3 DAX Measures

Avg Phosphorus = AVERAGE(WaterQuality[Phosphorus])

Avg DO = AVERAGE(WaterQuality[Dissolved_Oxygen])

Avg Turbidity = AVERAGE(WaterQuality[Turbidity])

VISUALIZATION DESIGN

The dashboard integrates multiple visualization techniques:

- KPI Cards → environmental status indicators
 - Scatter Plot → Rainfall vs Turbidity correlation
 - Bar Charts → Land Use comparison
 - Slicers → interactive filtering (Date, Station, Land Use)
-

KEY FINDINGS

- Rainfall is positively correlated with turbidity, indicating runoff-driven sediment transport.
 - Agricultural land use is the primary contributor to nutrient pollution.
 - Urban areas show moderate but persistent contamination.
 - Forest areas act as natural buffers maintaining water quality stability.
 - Pollution events significantly reduce fish population levels.
-

DISCUSSION

Water quality degradation is primarily driven by land-use intensity and rainfall variability.

Agricultural runoff is the dominant source of nutrient pollution, while rainfall intensifies pollutant transport.

These interactions highlight the importance of integrated watershed management strategies combining land-use regulation and hydrological monitoring.

LIMITATIONS

- Limited temporal coverage of dataset
- Missing industrial discharge variables
- Correlation does not imply causation

- Seasonal variation not fully captured
-

RECOMMENDATIONS

- Implement agricultural buffer zones to reduce runoff
 - Improve urban stormwater management systems
 - Increase monitoring frequency during rainfall events
 - Focus interventions on high-risk pollution stations
 - Integrate land-use planning with watershed protection policies
-

CONCLUSION

This study demonstrates that water quality is primarily influenced by land-use intensity and rainfall-driven runoff.

Agricultural activities are the dominant contributor to nutrient pollution, while forest ecosystems provide natural stabilization.

The Power BI dashboard supports data-driven environmental decision-making and sustainable watershed management.

APPENDIX

Include:

- Dashboard screenshots
- KPI cards
- Scatter plots
- Land-use comparison charts